

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 1

ADDITION OF TWO 8-BIT NUMBERS

AIM: To write and execute an 8085 assembly language program to add two 8-bit numbers and store the result in memory.

ALGORITHM

1. Load the first number from memory location **2500H** into the accumulator.
2. Move the first number to register **B**.
3. Load the second number from memory location **2501H** into the accumulator.
4. Add the contents of register B to the accumulator.
5. Store the sum in memory location **2502H**.
6. If there is a carry, store **01H** in memory location **2503H**; otherwise store **00H**.
7. Stop the program.

CODE:

```
LDA 2500H
MOV B,A
LDA 2501H
ADD B
STA 2502H
HLT
```

OUTPUT:

